



# AXH air-coolers

## air-x-limited

401 E. Lowry Road Phone (918) 283-9200  
Claremore, OK 74017 Fax (918) 283-9229  
[info@axh.com](mailto:info@axh.com) [www.axh.com](http://www.axh.com)

**Proposal / Job No.** 236449  
**Date** 11/9/2023  
**Page** 1 of 1

|   |               |                   |                  |                 |
|---|---------------|-------------------|------------------|-----------------|
| 1 | Purchaser     | PROPAK SYSTEM LTD | Ultimate User    | HESS            |
| 2 | Inquiry / PO# | W238195 / W238205 | Destination      | NORTH DAKOTA    |
| 3 | # Units       | 5                 | Model            | 156EF-R-112-26  |
| 4 | Assembly      | PACKAGED          | Draft            | FORCED          |
|   |               |                   | Reference        | EM2200HP-KBK4   |
|   |               |                   | Overall Size, Ft | Est Wt 40200 LB |

### THERMAL & MECHANICAL DESIGN

|    |                              |               |               |               |               |
|----|------------------------------|---------------|---------------|---------------|---------------|
| 5  | Service                      | IC-1          | IC-2          | IC-3          | AC            |
| 6  | Flow                         | 10.2MMSCFD    | 10.3MMSCFD    | 10.2MMSCFD    | 9.3MMSCFD     |
| 7  | Fluid                        | .8493SPGR     | .8505SPGR     | .8493SPGR     | .8502SPGR     |
| 8  | Temp. In / Out, °F           | 203.0 / 123.0 | 231.0 / 120.0 | 232.0 / 120.0 | 244.0 / 115.0 |
| 9  | Pressure, PSI                | 86PSIG        | 205PSIG       | 453PSIG       | 1100PSIG      |
| 10 | Pressure Drop, PSI           | 1.4           | 3.9           | 6.5           | 5.8           |
| 11 | Heat Load, BTU/HR            | 1100386       | 1604500       | 1759500 *     | 2148400 *     |
| 12 | True LMTD                    | 39.9          | 40.0          | 36.4          | 40.3          |
| 13 | Overall Rate, U              | 56.6          | 72.6          | 92.2          | 93.9          |
| 14 | Fouling Factor               | .0020         | .0020         | .0020         | .0020         |
| 15 | Surface, Tube / Total, Sq Ft | 492 / 10531   | 560 / 11974   | 526 / 11253   | 758 / 11300   |
| 16 | Sections, #                  | (1)           | (1)           | (1)           | (1)           |
| 17 | Design Temp, °F Max / Min    | 350 / -40     | 350 / -40     | 350 / -40     | 350 / -40     |
| 18 | MWP / Test Press, PSIG       | 215 / 280     | 645 / 839     | 645 / 839     | 1650 / 2145   |
| 19 | Pass Arrangement             | CROSSFLOW     | CROSSFLOW     | CROSSFLOW     | CROSSFLOW     |
| 20 | # Tube Rows                  | 5             | 5             | 5             | 4             |
| 21 | # Tube Passes                | 1             | 2             | 3             | 4             |
| 22 | Tubes, OD x BWG              | 1X16          | 1X16          | 1X16          | 3/4X16        |
| 23 | Material                     | SA179 STEEL   | SA179 STEEL   | SA179 STEEL   | SA249TP316LSS |
| 24 | # Per Section / Length, Ft   | 73 / 26       | 83 / 26       | 78 / 26       | 150 / 26      |
| 25 | Turbulators                  |               |               |               |               |
| 26 | Accelerators                 |               |               |               |               |
| 27 | Fins, Type                   | HI-EFF        | HI-EFF        | HI-EFF        | HI-EFF        |
| 28 | Material                     | AL            | AL            | AL            | AL            |
| 29 | Nozzles, Rating / Type       | 150RF         | 300RF         | 300RF         | 900RF         |
| 30 | Material                     | SA350LF2      | SA350LF2      | SA350LF2      | SA350LF2      |
| 31 | #Inlets / Size In            | (1) 8         | (1) 6         | (1) 4         | (1) 4         |
| 32 | #Outlets / Size In           | (1) 8         | (1) 6         | (1) 4         | (1) 4         |
| 33 | Headers, Type                | BOX           | BOX           | BOX           | BOX           |
| 34 | Material                     | SA516-70N     | SA516-70N     | SA516-70N     | SA516-70N     |
| 35 | Corrosion Allow, In          | .1250         | .1250         | .1250         | .1250         |
| 36 | Grooved Tubesheet            | DBL           | DBL           | DBL           | DBL           |
| 37 | Plugs, Type                  | SHOULDER      | SHOULDER      | SHOULDER      | SHOULDER      |
| 38 | Plugs Material               | SA350LF2      | SA350LF2      | SA350-LF2     | SA350LF2      |
| 39 | PWHT                         | YES           | YES           | YES           | YES           |
| 40 | ASME Code & Nat'l Board      | YES           | YES           | YES           | YES           |
| 41 | CRN                          | AB            | AB            | AB            | AB            |
| 42 | Add'l Specs & Options        |               |               |               |               |
| 43 | API                          | 661**         | 661**         | 661**         | 661**         |
| 44 | Louvers                      | AUTO /        | AUTO /        | AUTO /        | AUTO /        |
| 45 | Hail Screen                  |               |               |               |               |
| 46 | Inspection / NDT             | BX, UT, H     | BX, UT, H     | BX, UT, H     | BX, UT, H     |

| AIR-SIDE PERFORMANCE |                         |                           | FAN DATA           |                    | DRIVER DATA                | STRUCTURAL              |             |
|----------------------|-------------------------|---------------------------|--------------------|--------------------|----------------------------|-------------------------|-------------|
| 47                   | Ambient Air Temp, In °F | 100                       | Fan(s)             | (1) MOORE-10K-48EC | Type                       | Guards                  | FAN / DRIVE |
| 48                   | Elevation, Ft           | 2382                      | Blade Material     | ALUMINUM           | V-BELT DRIVE BY (1) 60HP,  | SOFT BUGSCREEN          |             |
| 49                   | Air Flow, SCFM          | 203650                    | HP / Fan           | 53.4               | 1200RPM, 460/60/3, IEEE, □ | AUTOMATIC WARM AIR      |             |
| 50                   | Outlet Air Temp, °F     | 129.9                     | Dia, In / # Blades | 156 / 10           | VFD COMP. MOTOR (CSA)      | RECIRCULATION SYSTEM    |             |
| 51                   | Min Air Temp, °F        | -40                       | RPM                | 237                | IEEE-841                   | WITH INTEGRAL HAILGUARD |             |
| 52                   |                         |                           | Tipspeed, FPM      |                    | 9679                       |                         |             |
| 53                   | Est. Noise Data:        | 84 dBA @ 1m, 66 dBA @ 15m | Pitch, Deg         |                    | 11.3                       | MURPHY VIBRATION SWITCH |             |

|    |                                                                                         |  |
|----|-----------------------------------------------------------------------------------------|--|
| 54 | Additional Info.                                                                        |  |
| 55 | * - INCLUDES 75 LB/HR AND 34 LB/HR CONDENSING IN THE IC-3 AND AC SERVICES RESPECTIVELY. |  |
| 56 | ** - IN ACCORDANCE WITH AXH EXCEPTIONS AND CLARIFICATIONS                               |  |
| 57 |                                                                                         |  |
| 58 |                                                                                         |  |
| 59 |                                                                                         |  |

|            |                                                                                    |                                          |
|------------|------------------------------------------------------------------------------------|------------------------------------------|
| NDT Legend | FX= 100% X-Ray of all header seam & nozzle butt welds with UT of attachment welds. |                                          |
|            | SX= Spot X-Ray of 1 long seam & 1 end closure, per header.                         | BX= 100% X-Ray of all nozzle butt welds. |
|            | UT = 100% UT of all header seam and nozzle attachment welds.                       | H = Hardness testing.                    |